



Assistant Manager

ARUN KUMAR CHAUBEY

Data Scientist

Contact

+91-9773862482
chaubeyarun.arun@gmail.com
Pune, India
linkedin.com/in/arun-k-chaubey

Impact at a glance

75%
less claim-review workload (100 → 25 claims reviewed)

3.2x
fraud yield per investigation (10% → 32% hit rate)

50K claims
claim-level fraud feature store, 60+ features from 13 datasets

+10%
revenue from sales, churn & lead-conversion models

64% → 98%
data quality after ETL & automation overhaul

Core skills

PROGRAMMING & DATA

Python Pandas NumPy
Scikit-learn SQL SQL Server
SAS

MACHINE LEARNING

Predictive Modeling Regression
Classification Clustering
Anomaly Detection Statistical Analysis
Decision Trees Random Forest
Gradient Boosting LightGBM
CatBoost

AUTOML & EXPLAINABILITY

FLAML Optuna
Hyperparameter Tuning Calibration
PR-AUC Lift & Gain SHAP

FRAUD & PRIVACY

Fraud Analytics Synthetic Data
PII Masking

GENAI & OCR

Ollama Qwen 2.5 EasyOCR

BI & ETL

Power BI DAX Power Query
SSIS

Education & Languages

B.Tech / B.E.
Electronics / Telecommunication
Pranveer Institute of Technology, Kanpur

Languages: English, Hindi

PROFILE

Data Scientist with 7 years in data science and analytics engineering, specializing in fraud analytics, propensity modeling and forecasting. I have led strategic planning initiatives and turn messy data into production-ready models and BI that move the business: data quality lifted from 64% to 98%, a 10% revenue increase and a 15% gain in sales productivity. Hands-on from synthetic-data generation, AutoML and explainable AI (SHAP) to locally hosted LLM pipelines, Power BI and ETL automation. I also spearhead training programs and a culture of continuous learning.

EXPERIENCE

Assistant Manager (Data Scientist)

Oct 2024 – Present

EXL

CLIENT PROJECT UK insurance client: claims fraud & anomaly detection

- **Business impact in claims triage:** before the model, investigators reviewed 100 claims to find 10 frauds; after it, they reviewed only the **25 highest-risk claims and found 8**, a **75% reduction in review workload** and a **3.2x higher fraud yield** per investigation (10% to 32%). FNOL handlers now focus on high-risk claims, with potentially lower operating cost.
- Evaluated fraud and anomaly-detection models with imbalance-aware metrics: **PR-AUC, probability calibration, cumulative gain and lift**.
- Delivered **explainable AI with SHAP**: global feature importance plus claim-level explanations for transparent fraud-risk evaluation.
- Used **MLOps in Microsoft Fabric** to select the best model, then save and deploy it.
- Automated ingestion, ETL, feature engineering and statistics (Python, SQL, SAS) to speed up model development; built Power BI dashboards (DAX, Power Query) for real-time claims and underwriting-risk monitoring.

IN-HOUSE PROJECT Fraud Shield

- Built an end-to-end **insurance fraud-detection pipeline**: consolidated 13 relational datasets (claims, vehicles, claimants, policy history, liability, suspicious agents, MIAFTR, Capstone, geographic risk) into a claim-level feature store of **50,000 claims and 60+ engineered features** (temporal, behavioral, relational).
- Ran EDA, then detected and removed **target leakage** (permutation importance, after an unrealistic F1 of 1.00) and tuned models with **Optuna**.
- Selected models with **FLAML AutoML** across LightGBM, CatBoost, Random Forest, Decision Trees and ensembles; **CatBoost** performed best on leakage-free features.
- Delivered **explainable AI with SHAP**: global feature importance plus claim-level explanations.
- **Project highlights:** a **synthetic motor-fraud data framework** (fraud rings, staged accidents, injected data-quality issues); a **PII-masking pipeline** for claim notes (regex + local Qwen 2.5 via Ollama, zero-hallucination replacement); a **policy-conversion propensity model** with percentile-based broker scoring; and **UK claims fraud risk scoring** that reduced false positives.

Manager (Data Scientist)

Jan 2023 – Dec 2023

NIBAV LIFTS PRIVATE LTD.

- Built ML models (Random Forest, Gradient Boosting, Regression, Decision Trees, GLM) for sales, churn and lead conversion, contributing to a **10% revenue**

increase; used GLMs to analyze business drivers and detect anomalies, and added a demand-forecasting model that cut inventory and production delays.

- Raised **data quality from 64% to 98%** with SSIS/SSDT ETL (Join, Lookup, Aggregate), SQL Server and Python automation.
- Designed Power BI dashboards (DAX, Power Query) for real-time sales, revenue, geo-spatial and operational KPIs; EDA and SQL analysis surfaced customer and regional sales patterns, shared with sales, operations and marketing.

Manager (Data Analyst & Scientist)

Feb 2020 – Jul 2022

BYJU'S

- Built predictive models (Regression, Decision Trees) for sales performance, lead conversion and engagement, plus a **student learning-path recommender** that boosted engagement and upsell.
- SQL funnel analysis to find top channels, plan territories and prioritize leads: **+15% sales productivity**; redesigned schemas and automated SSIS ETL, **cutting reporting delays by 30%**.
- Power BI (DAX, Power Query) dashboards for funnel health, campaign ROI, subscriptions and rep performance; Python automation freed **25% of analyst time**; insights for campaigns, pricing and course bundling.

Executive Data Analyst

Jan 2019 – Feb 2020

EXTRAMARKS EDUCATION

- SQL and Python for customer segmentation and predictive analytics: **+15% revenue**; cut Power BI refresh time by **70% (15 to 4.5 min)**; worked with global teams to raise operational efficiency by 20%.

SELECTED PROJECTS

Synthetic Motor-Insurance Fraud Data Framework

PYTHON · SYNTHETIC DATA · FEATURE ENGINEERING

Production-ready generator of multi-table relational data (claims, vehicles, claimants, policy history, liability, suspicious agents, external fraud sources) modeling fraud rings, staged accidents, identity misrepresentation and entity reuse. Injects realistic data-quality issues and validates business rules; produces privacy-compliant data for preprocessing validation and end-to-end ML pipeline testing, with aggregated and network-based fraud features.

PII Masking Pipeline for Claim Notes

PYTHON · REGEX · OLLAMA (QWEN 2.5) · PANDAS · MULTIPROCESSING

Hybrid design: deterministic regex for structured PII and a locally hosted LLM that returns names and addresses as JSON, followed by exact string replacement in Python for a **zero-hallucination** guarantee. Multi-worker execution with timeouts and review flags handles large Excel datasets, enabling secure data sharing for analytics and ML.

Fraud Detection & Claims Risk Analytics (UK Insurance)

REGRESSION · DECISION TREES · POWER BI

Models on claimant demographics, accident patterns, agent behavior and fraud history; risk scores blending internal and external intelligence reduced false positives and enabled early intervention. Delivered a real-time fraud-monitoring dashboard for claims handlers and investigators.

Policy Conversion Propensity & Broker Scoring

PROPENSITY MODELING · POWER BI · DAX

Propensity model predicting whether a submission converts to an active policy (turnover, company age, broker success rate, premium size, claim history, jurisdiction), plus a percentile-based broker-ranking framework. Delivered Power BI dashboards for senior management across claims, fraud, underwriting conversion and broker performance.